

# ELENA HERTEL

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## Summary

Software engineer with experience building end-to-end, product-facing AI systems in fintech. Strong command of Python, backend APIs, and data pipelines; track record of partnering with non-technical stakeholders to turn ambiguous workflows into production tools.

## Education

### University of Chicago

Chicago, IL

B.S.: Majors - Computer Science & Biology

September 2020 - June 2024

GPA: 3.68/4.00

**Relevant Coursework:** Theory of Algorithms, Advanced Algorithms, Data Science for Computer Scientists

## Employment

### Morningstar

Chicago, IL

*Associate Software Engineer*

December 2024 - Present

- Working as a full-stack engineer on the Direct Advisory Suite Architecture team, building internal and client-facing AI applications used by financial advisors.
- Designing and building an internal AI-driven support automation system using LangGraph and MCP tools, integrating with Jira to accelerate L1–L3 ticket resolution and reduce manual triage.
- Owned improvements to a production RAG-based chatbot, increasing answer quality by ~20% through custom retrieval pipelines, prompt design, and automated evaluation, while balancing latency and reliability constraints.
- Wrote and maintained custom data ingestion pipelines to keep AI systems synchronized with evolving product documentation, ensuring freshness, traceability, and consistent behavior in production.

*Software Engineer - Part-time*

September 2024 - December 2024

- Designed and implemented an internal AI-assisted performance review tool, working closely with People & Culture and Legal teams to define compliant workflows and structured evaluation outputs.
- Built a reusable, JSON schema-driven form framework for chat-based AI interfaces, enabling structured user input and extending a previously chat-only platform for broader internal use.

### Human Computer Integration Lab, University of Chicago

Chicago, IL

*Researcher - Part-time*

February 2024 - December 2024

- Built an Apple Watch application in Swift to detect and classify habitual user behaviors using accelerometer and gyroscope data.
- Implemented data processing pipelines and trained activity classification models for future user studies.

### Hamilton Insurance Group

London, England

*Software Development Intern*

June 2023 - August 2023

- Contributed to the development of a GPT-based tool to automate insurance submission processing, used globally across the firm to improve operational efficiency and business throughput.
- Developed full-stack web components in React and backend Python services, including document preprocessing logic and unit tests integrated into the deployment pipeline.

### Blinklab

Princeton, NJ

*Clinical Research Associate - Part-time*

September 2022 - June 2023

- Led Chicago-based data collection efforts for a mobile health startup; performed data normalization and analysis in R to validate model performance and improve diagnostic accuracy.

## Skills

Languages: Fluent in English, German, and French

Technical Skills: Python, Java, C, React, R, Swift; experience with LangGraph, LangChain, REST APIs, data pipelines, and production AI systems